

AIDAN MCLEAN

Keswick, ON

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Education

McMaster University

September 2021 – April 2026

Bachelor of Applied Science, Honours Computer Science (Co-op)

Hamilton, ON

Relevant Coursework

- Data Structures
- Algorithms Analysis
- Software Design
- Functional Programming
- Computational Theory
- Database Management
- Computer Networks
- Project Management

Work Experience

McMaster University Faculty of Engineering

April 2026 – Present

Research Assistant II, Battery SOC Benchmark Platform

Hamilton, ON

- Architected and developed a cloud-hosted research platform on a Node.js/Express (TypeScript) REST API, enabling researchers to upload, benchmark, and compare battery State-of-Charge estimation models across standardized EV scenarios.
- Designed a PostgreSQL schema spanning authentication, model metadata, file storage, and evaluation results, governing submissions across the full data life cycle in accordance with research protocols.
- Built an asynchronous Docker-based evaluation pipeline that spawns an isolated container per submission, captures structured scoring output, and delivers results by automated email.
- Implemented cookie-based session authentication with BetterAuth, rate limiting, email verification, and role-based access control enforcing ownership-gated authorization on all protected routes.
- Built leaderboard and comparison visualizations surfacing performance trends across 18+ evaluation metrics, and deployed the production environment with Docker Compose, NGINX, PM2, and SSL/TLS.

Regional Municipality of York

May 2025 – September 2025

Software Developer Co-op Student

Newmarket, ON

- Customized Drupal Webforms with JavaScript to implement dynamic form behavior, validation, and improved user experience.
- Built front-end integrations with Microsoft APIM to apply business logic, route requests securely, and connect forms to backend services.
- Developed a TypeScript front-end application that integrated with an internal GIS map server to retrieve and visualize location-based data.

Regional Municipality of York

May 2022 – September 2022; May 2024 – September 2024

Customer Service System Support Analyst Student

Newmarket, ON

- Effectively analyzed, diagnosed, and resolved technical issues for clients with varying knowledge and experience levels using diagnostic techniques and relevant questions.
- Logged precisely detailed tickets to ensure readability and clarity.
- Answered incoming calls, messages, and emails in a timely and efficient manner, prioritizing tasks based on urgency and impact.
- Installed and configured a variety of software for end users.

Regional Municipality of York

May 2023 – December 2023

Transit Fleet Analyst Co-op Student

Richmond Hill, ON

- Streamlined operations by designing and implementing Excel macros for task and process automation.
- Wrote, deployed, and maintained a PowerShell script for daily analysis of fueling data, enabling early detection and resolution of issues in transit fleet operations.
- Resolved a variety of technical issues for YRT Fleet staff and end users, including those with M5, Citrix, and Office.
- Conducted comprehensive and meticulous testing for M5 v23, ensuring robustness and optimal functionality prior to release.

Relevant Projects

Endnote Records

July 2026 – Present

E-Commerce Platform, Sole Developer

Remote

- Designed and deployed a bespoke two-application e-commerce platform for an independent record label: a public Next.js 16 storefront and a role-gated admin dashboard, sharing a 36-table Postgres schema, auth layer, and component library across a pnpm and Turborepo monorepo of six workspaces.
- Built a checkout that reprices every line server-side from the database, reserves stock inside a single transaction using deterministically ordered row locks, and releases reservations through an atomic guarded state transition shared by the webhook, expiry, and cron paths.
- Made Stripe webhook handling idempotent by persisting raw events, locking the event row before processing, and keying order creation on the session ID, with subtotal and currency mismatch detection that flags divergent orders for human review instead of failing silently.
- Implemented dual USD and CAD pricing with no FX conversion, server-rendering both amounts and selecting between them with a pre-paint script, which kept product pages on static generation with ISR rather than deopting to dynamic rendering.
- Enforced authorization at the permission level with better-auth's access-control plugin rather than role flags, validated every server action with Zod, and wrote an append-only audit log capturing actor, action, and before/after values for each mutation touching money, inventory, or user data.
- Concentrated 101 Vitest unit tests on pricing resolution, shipping quotes, discount eligibility, and permissions, gated behind a single build pipeline, with GitHub Actions applying Neon migrations and running scheduled jobs.

Technical Proficiencies

Languages: TypeScript/JavaScript, Python, Java, SQL, C, Bash, PowerShell

Frontend: React, Next.js (App Router, Server Components), Tailwind CSS, shadcn/ui, Base UI, responsive design

Backend & Data: Node.js, Express, REST API design, PostgreSQL, Drizzle ORM, schema design, Zod, better-auth, session auth and RBAC, Stripe

Infrastructure & DevOps: Docker, Docker Compose, NGINX, PM2, SSL/TLS, Linux, OpenStack, Vercel, Neon, GitHub Actions CI/CD

Testing & Tooling: Vitest, Git, pnpm, Turborepo, ESLint, Prettier, Agile/Scrum

Enterprise Platforms: Drupal, Microsoft APIM, Power Automate, Azure Entra, GIS services